

Brazilian Mathematical Community: a brief history

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Abstract

This article examines the historical processes that led to the emergence, consolidation, and internationalisation of the Brazilian mathematical community during the twentieth century. Rather than treating this development as the cumulative result of individual achievements, the analysis adopts an institutional and sociological perspective, focusing on the mechanisms through which a coherent and durable scientific community was formed. Three interrelated dimensions structure the discussion: the establishment of institutions dedicated to training mathematicians and supporting research; the creation of national information and communication infrastructures; and the circulation of people, ideas, and practices through international exchanges. University-based mathematics programmes created in the 1930s in São Paulo and Rio de Janeiro addressed the foundational problem of training and reproducing researchers. From the late 1950s onward, the Brazilian Mathematics Colloquia played a central role in consolidating the community by providing regular national forums for intellectual exchange, standardising research expectations, and linking mathematicians across regions and generations. Research institutes such as the Institute of Pure and Applied Mathematics (IMPA) contributed to this process by strengthening postgraduate training and international integration. By situating these developments within a broader comparative and international context, the article argues that the Brazilian case illustrates how mathematical communities in the Global South are actively constructed through sustained institutional investment, strategic integration into global networks, and the development of robust communication infrastructures. In this light, later markers of international recognition—such as Brazil’s position within the International Mathematical Union and individual distinctions—are best understood as outcomes of long-term structural processes rather than isolated successes.

1 Introduction

Brazil’s university system is relatively recent. Institutions of higher education aimed at training specialised professionals—such as medical schools and the Royal Military Academy—were created only after the arrival of the Portuguese royal family in 1808. Throughout the imperial period (1822–1889) and the Old Republic (1889–1930), scientific institutions did emerge, but they largely followed a pragmatic stance, oriented toward immediate professional and technical needs rather than sustained scientific research [Sil03].

In this context, mathematics occupied a marginal and instrumental position. Mathematical training was primarily embedded within military and engineering schools, most notably the Royal Military Academy, whose curriculum devoted its first four years to mathematics to train artillery officers, engineers, geographers, and topographers. Although degrees and even doctorates in Mathematical Sciences were formally regulated in the mid-nineteenth century, and a small number of theses were defended, the broader conditions necessary for the development of a research-oriented mathematical culture were absent. Mathematical work remained largely isolated, dependent on individual effort and self-education, and weakly connected to international scientific circuits [Sil03, AS11].

Until the early twentieth century, engineering schools functioned as the principal sites for the dissemination of mathematical knowledge in Brazil. This situation began to change only in the 1930s, with the creation of university courses specifically dedicated to mathematics and no longer subordinated to engineering training. The establishment of the Faculty of Philosophy, Sciences and Letters at the University of São Paulo (FFCL-USP, from *Faculdade de Filosofia, Ciência e Letras, Universidade de São Paulo*) in 1934 and the School of Sciences at the Federal District University

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(UDF, from *Universidade do Distrito Federal* then in Rio de Janeiro) in 1935 marked a decisive institutional shift. These initiatives created, for the first time, stable environments for advanced mathematical training and research, enabling the formation of research groups, the supervision of doctoral work, and sustained interaction with foreign mathematicians [Sil03, d'A08].

Over the following decades, particularly in the 1940s and 1950s, the number of mathematics courses, trained researchers, and research institutions increased significantly across different regions of Brazil. New communication channels—journals, seminars, scientific societies, and national meetings—were created, allowing Brazilian mathematicians to exchange ideas, coordinate research agendas, and collectively define standards of training and scientific practice. Amongst these, special emphasis should be given to the Brazilian Mathematical Colloquium and the Brazilian Mathematical Society. In parallel, funding agencies such as CNPq and CAPES, founded in the early 1950s, provided material support for research, postgraduate training, and international exchange. These developments laid the groundwork for the emergence of what can be meaningfully described as a Brazilian mathematical community.

From the 1950s onward, the Brazilian mathematical community entered a period of consolidation and gradual internationalisation. This development was supported by the expansion of university departments, the creation of research institutes, and the establishment of national channels for scientific communication. The Brazilian Mathematics Colloquium (Colóquio Brasileiro de Matemática), first held in 1957, became a key forum for bringing together mathematicians from different regions of the country and for disseminating contemporary mathematical developments. The creation of the Brazilian Mathematical Society (Sociedade Brasileira de Matemática – SBM) in 1969 further strengthened these networks by promoting scientific meetings, publications, and institutional coordination within the field. Research institutes—most notably the Institute of Pure and Applied Mathematics (IMPA), founded in 1952—also played an important role in fostering research and international collaboration during this period. In recent decades, this long-term process has culminated in internationally recognised achievements, including Brazil's inclusion in Group 5 of the International Mathematical Union, the hosting of the International Congress of Mathematicians in Rio de Janeiro in 2018, and the award of the Fields Medal to Brazilian mathematician Arthur Ávila in 2014.

Against this background, and drawing on recent historiographical work that emphasises the historically situated nature of national scientific communities and the role of educational systems, institutions, and communication infrastructures in their formation ([Sch21]), this article examines the historical processes that led to the creation, consolidation, and internationalisation of the Brazilian mathematical community. The analysis focuses on three interrelated dimensions: the establishment of institutions dedicated to training mathematicians and hosting research; the construction of national communication channels, such as journals, societies, and scientific meetings, and the circulation of people, ideas, and practices through international exchanges. By tracing these developments from the 1930s to the late twentieth century, the article seeks to clarify how mathematics in Brazil moved from a fragmented and peripheral activity to a nationally organised and internationally recognised scientific community.

2 Mathematical Communities and Their Formation

The notion of a mathematical community employed in this article is not merely descriptive, nor does it refer simply to the coexistence of individuals engaged in mathematical activity within a given territory. Rather, it designates a historically situated social formation, characterised by relatively stable institutions, shared norms of training and evaluation, recognised channels of communication, and sustained interaction—both internal and international—among its members.

Following Thomas Kuhn's classic characterisation of scientific communities[Kuh62], a mathematical community can be understood as a group whose members share common standards of professional competence, a corpus of problems regarded as legitimate, and criteria for what counts as an acceptable solution. Crucially, such communities are reproduced through education and training, particularly at the postgraduate level, and through participation in collective practices such as seminars, conferences, and peer review. From this perspective, the emergence of a mathematical community presupposes not only individual mathematical production but also the existence of social and institutional mechanisms that enable continuity across generations.

Sociological and historical studies of science further emphasise the role of institutions and communication infrastructures in stabilising scientific communities. As Schwartzman [Sch01] and Bastos [Bas06] have argued, research communities require institutional spaces that support full-time scientific work, mechanisms for professional recognition, and creation and access to material

resources. Equally important are channels through which knowledge circulates—journals, textbooks, learned societies, and scientific meetings—which allow dispersed individuals to recognise themselves as part of a collective enterprise and to coordinate their activities.

The formation of a national mathematical community cannot be reduced to the appearance of isolated researchers or sporadic publications. It depends instead on the successful articulation of training institutions, research centres, and communication networks capable of integrating local mathematical activity into broader international circuits.

This article therefore treats the Brazilian mathematical community as the outcome of a historical process rather than as a given entity. Analytically, three interrelated dimensions guide the discussion that follows. First, the creation of institutions dedicated specifically to the training of mathematicians and the support of research, particularly universities and research institutes. Second, the circulation of people, ideas, and practices through international exchanges, visiting professorships, and postgraduate training abroad. Third, the construction of national communication infrastructures—such as journals, scientific societies, and regular meetings—that enabled Brazilian mathematicians to share work, establish common standards, and develop a sense of collective identity.

By adopting this framework, the subsequent sections move beyond a purely chronological or biographical account. Instead, individual actors, institutions, and events are analysed in terms of the roles they played within these broader processes of community formation, consolidation, and internationalisation.

3 First courses in Mathematics

The creation of the University of São Paulo (USP) in 1934 represents a decisive moment in the history of mathematics in Brazil, not merely because it introduced new degree programmes, but because it established an institutional environment explicitly designed to sustain scientific research on a continuous basis. In contrast to earlier professional schools, USP was conceived as a research-oriented university, and its Faculty of Philosophy, Sciences and Letters (FFCL) was entrusted with the task of cultivating the basic sciences, including mathematics, as autonomous fields of inquiry.

The mathematics section of the FFCL-USP embodied a deliberate institutional strategy. Rather than relying solely on locally trained engineers or military instructors, the university recruited foreign mathematicians who were active participants in contemporary research traditions. A key feature of the USP project was the deliberate recruitment of foreign mathematicians active in contemporary research traditions. Italian scholars played an important role in the 1930s, and French mathematicians became increasingly influential from the mid-1940s onward. Figures such as Luigi Fantappiè, Giacomo Albanese, and especially Dieudonné, Zariski, Weil, and Grothendieck (even when present only briefly or indirectly) played a crucial role in introducing modern mathematical perspectives, research-oriented curricula, and seminar-based teaching practices. Their presence contributed not only to the transmission of advanced content but also to the establishment of norms regarding what it meant to do mathematics as a scientific profession.

Equally important was the creation of stable conditions for training new generations of mathematicians. At USP, mathematics was no longer an auxiliary component of engineering education but the core of a dedicated degree programme, structured to lead students progressively from foundational coursework to advanced topics and, eventually, to original research. Seminars, guided reading of recent international literature, and supervision of student work fostered forms of collective practice that are characteristic of functioning research communities.

The impact of USP extended well beyond the city of São Paulo. Graduates trained at the FFCL-USP went on to occupy teaching and research positions in other emerging centres, contributing to the diffusion of research-oriented mathematical practices throughout the country. Through this process, USP functioned as a nodal institution within a nascent national network, linking local initiatives to international mathematical circuits and providing a reference model for subsequent university-based mathematics programmes.

From the perspective adopted in this article, the importance of USP lies less in the individual achievements of particular mathematicians—significant though these were—than in the institutional configuration it established. By combining dedicated degree programmes, full-time academic positions, exposure to international research cultures, and collective forms of intellectual work, USP created the conditions under which mathematics in Brazil could begin to operate as a coherent scientific activity. It thus constitutes a relevant element in the historical process that led to the emergence of a Brazilian mathematical community.

While the University of São Paulo provided an early and influential model for the institutionalisation of mathematical research, the development of mathematics in Rio de Janeiro followed a distinct but complementary trajectory. As the former federal capital and a longstanding centre of higher education and scientific administration, Rio hosted a dense constellation of institutions whose interaction shaped a different pattern of community formation—one characterised less by a single dominant university and more by institutional plurality.

The creation of the University of the Federal District (UDF) in 1935, and later the incorporation of its mathematics activities into the National Faculty of Philosophy (FNF_i) of the University of Brazil, represented an important attempt to establish university-based mathematics along lines comparable to those pursued in São Paulo [Car20, d'A08]. As in the case of USP, the introduction of dedicated mathematics courses, the recruitment of foreign professors, and the gradual separation of mathematics from engineering curricula created new spaces for advanced training and research. However, political instability and institutional discontinuity limited the capacity of these initiatives to generate the same degree of long-term stability achieved at USP.

Despite these constraints, Rio de Janeiro played a crucial role in sustaining and diversifying Brazilian mathematical activity. The presence of multiple higher education institutions—including FNF_i, military schools, and later private universities—allowed mathematics to develop across overlapping but partially autonomous settings. This multiplicity fostered a broader base of teaching and contributed to the dissemination of mathematical training, even when research activity remained more fragmented.

From the 1940s onward, Rio also became increasingly important as a centre of scientific policy and research administration. The establishment of institutions such as the Brazilian Center for Research in Physics (CBPF) and, subsequently, IMPA, anchored high-level research in the city and altered its mathematical landscape decisively. In this sense, Rio's significance for the Brazilian mathematical community does not lie solely in its early university initiatives, but in its later function as a site where research institutes, funding agencies, and national scientific organisations converged.

Private institutions, amongst others, the Pontifical Catholic University of Rio de Janeiro (PUC-Rio), further contributed to this diversified ecosystem. Although smaller in scale than major public universities, PUC-Rio developed strong programmes in mathematics and related areas, particularly through close connections with IMPA and an emphasis on research-oriented postgraduate training. Such institutions exemplify how, by the second half of the twentieth century, the Brazilian mathematical community had become sufficiently consolidated to support multiple institutional forms beyond the original public university model.

Analytically, Rio de Janeiro illustrates a complementary mode of community formation. Whereas São Paulo functioned as an early stabilising nucleus for university-based mathematics, Rio evolved into a plural and strategically significant centre in which teaching institutions, research institutes, and national organisations intersected. This configuration proved essential for the later consolidation and internationalisation of Brazilian mathematics, reinforcing the community's institutional depth and geographical reach.

Although the institutions in the cities of São Paulo and Rio de Janeiro have been highlighted for their pioneering spirit and relevance to the more modern training of Brazilian mathematicians, it should be noted that until the end of the 1950s, mathematics courses also existed in the capitals of the states of Rio Grande do Sul, Bahia, Minas Gerais, Paraná, and Pernambuco, in addition to private institutions in the city of São Paulo and in Campinas – SP, and public institutions in Rio Claro – SP. It is also noteworthy that research groups were formed in some of these institutions and in others such as ITA and the School of Engineering of São Carlos (USP – São Carlos).

4 Research Institute: IMPA

The founding of the Institute of Pure and Applied Mathematics (IMPA) in 1952 marks a second, qualitatively distinct phase in the formation of the Brazilian mathematical community. Whereas the creation of university mathematics programmes in the 1930s addressed the problem of initial training and the introduction of research practices, IMPA responded to a different set of structural needs: the consolidation of high-level research, the coordination of national efforts, and the sustained integration of Brazilian mathematics into international scientific networks.

IMPA emerged in a context shaped by broader postwar transformations in Brazilian science policy, particularly the establishment of national funding agencies such as the National Research Council (CNPq) and the Coordination for the Improvement of Higher Education Personnel (CAPES). Unlike university departments, which were embedded within teaching-oriented institutional frame-

works, IMPA was conceived as a research institute with a clear mandate to prioritise scientific production, advanced training, and international collaboration. This institutional autonomy allowed it to operate as a focal point for mathematical research at the national level [Dyn09].

Over the years, at IMPA, stable conditions for postgraduate education in mathematics were created. By concentrating senior researchers, postdoctoral fellows, and graduate students within a single institutional space, IMPA enabled forms of intensive intellectual exchange. Regular seminars, visiting professorships, and long-term research programmes fostered a collective research culture and helped standardise expectations regarding research quality, doctoral training, and professional advancement. IMPA also played an important role in articulating the Brazilian mathematical community internally. Through its involvement in the organisation of the Brazilian Mathematics Colloquia and its close relationship with the Brazilian Mathematical Society (SBM), the institute contributed to the creation of national communication infrastructures. These initiatives facilitated interaction among mathematicians from different regions, reduced institutional isolation, and reinforced a shared sense of belonging to a national scientific community.

At the same time, IMPA functioned as a privileged interface between Brazilian mathematics and the international mathematical world. The institute systematically invited foreign researchers, encouraged its members to undertake research stays abroad, and aligned its research agendas with internationally active areas—most notably in dynamical systems. This strategy proved decisive for the international visibility of Brazilian mathematics, leading to sustained participation in global research networks and, eventually, to internationally recognised achievements. By the early twenty-first century, the cumulative effects of these processes were reflected in Brazil’s position within international scientific governance. Its progression to Group 5 of the International Mathematical Union—entailing full membership and voting rights—can be read as an external indicator of institutional maturity, sustained research activity, and stable participation in global mathematical networks, rather than as a sudden or isolated achievement.

From an analytical standpoint, the importance of IMPA lies not simply in the scientific output of its members, but in its role as an institutional integrator. By concentrating resources for research, postgraduate training, and international exchange, IMPA played an important role in structuring Brazilian mathematical research and facilitating its international visibility. At the same time, this centralisation produced a highly hierarchical institutional landscape, in which IMPA occupied a particularly prominent position. As in other national scientific systems organised around leading research institutes, this configuration contributed both to the rapid internationalisation of the field and to persistent asymmetries between the institute and other university-based mathematics centres.

5 Communication infrastructure

In addition to universities and research institutes, the consolidation of the Brazilian mathematical community depended crucially on the development of information and communication infrastructures that enabled interaction, coordination, and collective self-recognition. Among these, the Brazilian Mathematics Colloquia (Colóquios Brasileiros de Matemática, CBMs) and the national mathematical olympiads played distinct but complementary roles in structuring the community across generations and institutional boundaries.

The Brazilian Mathematics Colloquia, first held in 1957 under the auspices of IMPA, constituted a decisive innovation in the national organisation of mathematical life. By providing a regular forum for extended lectures, research presentations, and informal exchange, the CBMs created a shared temporal and intellectual reference point for Brazilian mathematicians. Unlike sporadic conferences or isolated seminars, the colloquia were explicitly designed to bring together researchers from different regions, institutional backgrounds, and stages of career, thereby reducing fragmentation and fostering a sense of collective enterprise.

From an analytical perspective, the CBMs functioned as a central mechanism of community articulation. They contributed to the circulation of current research topics, helped standardise expectations regarding training and research quality, and facilitated the integration of younger mathematicians into national and international networks. The publication and dissemination of lecture notes associated with the colloquia further extended their impact, transforming ephemeral events into durable informational resources that shaped curricula and research agendas well beyond the meetings themselves.

The Brazilian Mathematical Society (SBM) was founded in 1969 at the seventh edition of the Brazilian Mathematical Colloquium (CBM) and initially had its headquarters at IMPA. This society facilitated scientific exchange among its members through the organization of scientific

meetings and the publication of the SBM Bulletin. Over the years, the SBM expanded its activities as the mathematical community grew. For example, the number of published journals increased, with different focuses; editorial lines for scientific manuals were established; the society expanded the events it supported; and the Mathematical Olympiads were created (Oliveira, 2018).

The Brazilian Mathematical Olympiad (OBM) and, later, the Brazilian Mathematical Olympiad for Public Schools (OBMEP) operated primarily at the pre-university level, identifying and nurturing mathematical talent on a national scale. By reaching students across diverse social and geographical contexts, these programmes broadened the recruitment base of the mathematical community and contributed to the early formation of mathematical interests and aspirations. These initiatives also formed part of a wider set of activities through which the Brazilian Mathematical Society (Sociedade Brasileira de Matemática – SBM) helped articulate the different layers of the mathematical system. Through its conferences, publications, outreach programmes, and involvement in olympiad initiatives, the SBM provided an institutional framework that connected professional research, university training, and school-level mathematical culture. In doing so, it reinforced shared standards of excellence and contributed to the long-term cohesion and visibility of the Brazilian mathematical community.

Taken together, the CBMs and mathematical olympiads illustrate how information and communication infrastructures operate across temporal scales. While the colloquia primarily structured interaction among established and emerging researchers, olympiads ensured the long-term renewal of the community by shaping future cohorts of mathematicians. Both mechanisms contributed to the consolidation of Brazilian mathematics by transforming dispersed individual activities into a recognisable, coordinated, and socially embedded scientific community.

6 Conclusion

This article has examined the emergence, consolidation, and internationalisation of the Brazilian mathematical community as a historically situated process, shaped by the interaction of institutions, people, and communication infrastructures rather than by the isolated achievements of individual mathematicians. By adopting an analytic notion of scientific community, the study has sought to move beyond a purely chronological or biographical narrative and to identify the mechanisms that enabled mathematics in Brazil to develop as a coherent and durable collective enterprise.

The analysis has shown that this process unfolded in distinct but interconnected phases. The creation of university-based mathematics programmes in the 1930s—most notably at the University of São Paulo and, in a different institutional configuration, in Rio de Janeiro—addressed the foundational problem of training mathematicians within a research-oriented framework. These initiatives introduced new curricula, pedagogical practices, and professional norms, making possible the systematic reproduction of researchers for the first time. However, on their own, universities were insufficient to ensure the long-term consolidation and coordination of mathematical research at the national level. A second phase, beginning in the 1950s, was marked by the establishment of IMPA and the expansion of national science policy through funding agencies such as CNPq and CAPES. IMPA functioned as an institutional integrator, concentrating research activity, structuring postgraduate education, and serving as a focal point for national and international exchange. Through its close links with journals, scientific societies, and national meetings, it contributed decisively to the articulation of a Brazilian mathematical community that was both internally connected and outward-looking.

Taken together, these developments highlight the importance of institutional diversity and complementarity. Rather than converging on a single model, Brazilian mathematics evolved through the interaction of different institutional forms—universities, research institutes, public agencies, and private institutions—each contributing in specific ways to training, research, and communication. This plural configuration proved crucial for the resilience and later expansion of the community, allowing it to absorb international influences while maintaining a national structure.

The Brazilian case also illustrates more general features of the formation of scientific communities in contexts often described as peripheral. It underscores the central role of circulation—of people, ideas, and practices—in mediating between local initiatives and international scientific standards, and it shows how sustained institutional investment is required to transform sporadic scientific activity into a stable research tradition. In this sense, the history of Brazilian mathematics offers a concrete example of how national scientific communities are actively constructed over time, rather than emerging spontaneously.

By the end of the twentieth century, the consolidation of institutions, training mechanisms,

and communication infrastructures had enabled Brazilian mathematics to achieve a level of international recognition that would have been difficult to envisage only a few decades earlier. More broadly, this study suggests that analysing scientific development through the lens of community formation provides a productive framework for understanding the dynamics of mathematical activity, both in Brazil and in other national contexts.

The award of the Fields Medal to Artur Avila in 2014 constitutes a particularly visible marker of this transformation. While individual in nature, such recognition presupposes the existence of a dense institutional and intellectual environment capable of supporting advanced research trajectories over extended periods. In this sense, the medal should be understood not as the culmination of a linear success narrative but as an outcome made possible by decades of investment in training, research infrastructure, and international integration.

7 Coda: BRICS and Mathematics

The Brazilian case also invites comparison with other large emerging economies, although a full comparative treatment lies beyond the scope of this article. The brief remarks that follow are intended only to situate the Brazilian trajectory within a broader research agenda on mathematics and scientific development in the Global South.

Beyond its national specificity, the Brazilian case¹ speaks to broader debates concerning the role of mathematics in the scientific and economic development of countries in the Global South. Mathematics occupies a distinctive position within the landscape of scientific knowledge: it underpins technological innovation, informs strategic decision-making across domains ranging from engineering to finance, and provides a shared language through which scientific communities participate in global exchanges. For this reason, many development strategies implicitly or explicitly assume that long-term economic resilience depends on the existence of a well-educated population capable not only of adopting imported technologies but also of adapting and innovating them in locally meaningful ways.

Within this perspective, the consolidation of a national mathematical community can be understood as a form of strategic intellectual investment. The Brazilian experience illustrates how sustained commitment to training, research infrastructure, and international integration can enable a country to reduce dependence on external expertise while remaining embedded in global scientific networks. This path has not been uniform across emerging economies. In comparison with other BRICS countries, Brazil followed a trajectory distinct from the long-established mathematical traditions of Russia and India, achieved a higher degree of institutional integration than South Africa, yet developed a community of more modest scale and influence than that of China. Such comparisons underscore that there is no single model for mathematical development: national histories, institutional choices, and geopolitical contexts shape different configurations of scientific capacity.

Seen in this light, the history of Brazilian mathematics is not only a case study in community formation but also a contribution to understanding how countries outside the traditional centres of scientific power can build durable, internationally connected research cultures. In this sense, mathematics—often perceived as abstract or detached from social concerns—emerges as a central component of broader projects of educational development, technological autonomy, and economic transformation in the Global South.

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